



Revision Question Bank

Semester: Fall 2023

Course Code: CS100X

Course Title: Introduction to IT for CS

Chapter: 01, 02, 03, 04, 05

2023/2024

Revision Question Bank

Answer the following MCQ questions

- Q1. Bit patterns are used to represent information such as _____.
(a) numbers
(b) characters
(c) images
(d) any of the above
- Q2. Boolean operation is an operation that manipulates _____.
(a) one or more T/F values
(b) one or more 1/0 values
(c) one or more true/false values
(d) any of the above
- Q3. Gate is a device that computes a Boolean operation.
(a) True
(b) False
- Q4. The output of the XOR gate is inverted in the NOR gate.
(a) True
(b) False
- Q5. The output of the NAND gate is always true if one of its inputs remains false.
(a) True
(b) False
- Q6. If the inputs of NAND gate are A & B, and the output is C, then the NAND gate can be described by _____.
(a) $C = (A.B)$
(b) $C = \text{Invert } (A+B)$
(c) $C = \text{Invert } (A.B)$
(d) $C = (A+B)$
- Q7. If the inputs of NOR gate are A & B, and the output is C, then the NOR gate can be described by _____.
(a) $C = (A.B)$
(b) $C = \text{Invert } (A+B)$
(c) $C = \text{Invert } (A.B)$
(d) $C = (A+B)$

Q8. Flip-flop is a circuit that maintains a state until directed by input to change the state.

- (a) True
- (b) False

Q9. Convert $(10110101)_2 = (\quad)_{16}$

- (a) 5B
- (b) 115
- (c) 181
- (d) B5

Q10. Convert $(329)_{10} = (\quad)_{16}$

- (a) 139
- (b) 194
- (c) 149
- (d) 209

Q11. Convert $(23E)_{16} = (\quad)_{10}$

- (a) 512
- (b) 574
- (c) 48
- (d) 14

Q12. Convert $(F7D)_{16} = (\quad)_{10}$

- (a) 112
- (b) 13
- (c) 3965
- (d) 3840

Q13. Convert $(10011)_2 = (\quad)_{10}$

- (a) 3
- (b) 16
- (c) 19
- (d) Any of the above

Q14. Convert $(75)_{10} = (\quad)_2$

- (a) 01001100
- (b) 01001010
- (c) 01001000
- (d) 01001011

Q15. Convert $(11111110)_2 = (\quad)_{16}$

- (a) FE
- (b) EF
- (c) FD
- (d) DF

Q16. The most significant bit is the bit at _____.

- (a) at the right end
- (b) at the middle
- (c) at the left end
- (d) None of the above

Q17 The memory **address is a** name that uniquely identifies each cell in the computer's main memory.

- (a) True
- (b) False

Q18. Encoding information has been established for _____.

- a. numeric data
- b. images
- c. text
- d. all the above

Q19. Convert (Hello.) into ASCII = ()

- a. 72 101 108 108 111 46
- b. 73 102 109 109 112 46
- c. 71 100 107 107 110 46
- d. 72 101 108 108 111 45

Q20. Each pixel in RGB image can be saved into storage device using _____ byte(s).

- a. one
- b. two
- c. three
- d. four

Q21. Convert (1100)₂ = () 2's complement.

- a. 1001
- b. 0100
- c. 0111
- d. 1100

Q22. Convert (1101)₂ = () 1's complement.

- a. 1111
- b. 0000
- c. 0001
- d. 0010

Q23. $(0011) + (0010) = (\quad)_2$

- a. 0101
- b. 1000
- c. 1001
- d. 0011

Q24. $(7) + (-5) = (\quad)_2$

- a. 0001
- b. 0010
- c. 0011
- d. 0100

Q25. If adding two positive numbers (where $MSB = 0$) and the result ends up being negative ($MSB = 1$), then overflow occurred.

- (a) True
- (b) False

Q26. Each pixel in raw image can be saved into storage device using _____ byte(s).

- a. two
- b. three
- c. one
- d. four

Q27. if adding two negative numbers (where $MSB = 1$) and the result ends up being positive ($MSB = 0$), then once again overflow occurred.

- (a) True
- (b) False

Q28. If the five-bit representation 2's complement of two numbers (-10 & -7) are $-10 = 10110$ and $-7 = 11001$, then adding -10 and -7 will give _____.

- a. an overflow will occur,
- b. the result lies outside the range,
- c. the result equals 10001,
- d. any of the above

Q29. If a program is using an integer variable to store a number, but the number is too large to fit into an integer, an overflow error will occur.

- (a) True
- (b) False

Q30. A byte represents a single character in the computer.

- (a) False
- (b) True

Q31. Write a python statement to display Hello World!

- a. `print("Hello World!")`
- b. `cout << "Hello World!";`
- c. `print("Hello World!");`
- d. `print(Hello World!)`

Q32. The format of image compression is _____.

- a. GIF
- b. JPEG
- c. TIFF
- d. Any of the above

Q33. The format of audio is _____.

- a. MP3
- b. AAC
- c. FLAC
- d. Any of the above

Q34. The format of audio is _____.

- a. AVI
- b. MOV
- c. MP4
- d. Any of the above

Q35. The format of images is _____.

- a. PNG
- b. BMP
- c. JPG
- d. Any of the above

Q36The format of vector graphics is _____.

- a. EPS
- b. AI
- c. SVG
- d. Any of the above

Q37The format of raster graphics is _____.

- a. PSD
- b. PNG
- c. GIF
- d. Any of the above

Q38. An extra bit that is associated with the ASCII of letter C to make the total number of 1s odd, then the bit representation of letter C is _____.

- a. 101000000
- b. 01000001
- c. 101000001
- d. 001000011

Q39. With an associated odd parity check of 3-bit representation such as 1011 is transmitted, and the received bits are 1001, then _____.

- a. an error occurs.
- b. no error occurred.
- c. received bits should be 5 bits.
- d. None of the above

Q40. Which of the following gates gives an output zero, if all its inputs are the same.

- a. AND
- b. OR
- c. XOR
- d. NOT

Q41. Which of the following gates gives an output 1, if all its inputs are the same.

- a. AND
- b. OR
- c. NOT
- d. XNOR

Q42. With an associated odd parity check of 3-bit representation such as 1110 is transmitted, then the received bits are _____.

- a. 1001
- b. 1110
- c. 0011
- d. None of the above

Q43. Which data storage technology stores data temporarily, not permanently, for use by a device?

- a. DVD
- b. Hard drive
- c. USB memory
- d. RAM

Q44. Convert $(000)_2 = (\text{_____})_{10}$'s complement

- a. 1000
- b. 111
- c. 001
- d. 100

Q45. Convert $(000)_2 = (\text{_____})_1$'s complement

- a. 011
- b. 111
- c. 001
- d. 100

Q46. Addition using 2's complement $(1101 - 1110)_2 = (\text{_____})_2$

- a. 0001
- b. 1110
- c. 1111
- d. 0000

Q47. Addition using 2's complement $(-1101 - 1110)_2 = (\text{_____})_2$

- a. 00101
- b. 11011
- c. 11111
- d. 00000

Q48. Where do find computers?

- a. everywhere
- b. in cars
- c. airplanes
- d. space crafts

Q49. Data is _____

- a. information after process
- b. raw facts unusual for human beings.
- c. refined information
- d. refined knowledge

Q50. What is the system? The system is _____.

- a. a set of modules that work towards a common goal.
- b. a set of things working together as a mechanism or interconnecting network.
- c. input, process, and output.
- d. Any of the above

Q51. Personal computers maintain several kinds of computers such as _____.

- (a) notebook
- (b) tower computer
- (c) laptop
- (d) all the above

Q52. The most widely used handheld computers are _____

- (a) laptops and tablets
- (b) PDAs.
- (c) portable media players.
- (d) digital cameras

Q53. The word "software" is interchangeable with the word _____

- (a) procedure
- (b) application
- (c) program
- (d) computer

Q54. _____ is information after process.

- (a) Feedback
- (b) Knowledge
- (c) Output
- (d) Data

Q55. A computer contains many electronic and mechanical components known as _____.

- (a) CPU
- (b) Hard desk
- (c) Hardware
- (d) CD

Q56. A program that controls a computer's basic functions is called _____.

- (a) Hard Drive
- (b) Motherboard
- (c) Operating System
- (d) CPU

Q57. _____ is a special-purpose computer that functions as a component in a larger product.

- (a) Super computer
- (b) Mainframe
- (c) Embedded computer
- (d) Tablet

Q58. _____ send and receive data, instructions, and information between computers.

- (a) Input devices
- (b) Storage devices
- (c) Processing devices
- (d) Communication devices

Q59. _____ is best fastest computer in world that is very expensive.

- a. Embedded computer
- b. Supercomputer
- c. Tablet
- d. Microcomputer

Q60. The Internet is the largest _____ in the world.

- (a) system unit
- (b) network
- (c) database
- (d) operating system

Q61. System software includes all of the following except _____

- (a) operating systems
- (b) device drivers
- (c) utilities
- (d) browsers

Q62. Where do find computers? The answer is _____

- (a) everywhere
- (b) at the computers' labs in schools
- (c) at the organizations
- (d) at the banks

Q63. What is the system? The system consists of input, process, output, and a feedback from output to correct the input data.

- (a) False
- (b) True

Q64. A computer is an electronic device, operating under the control of instructions stored in its own memory.

- (a) True
- (b) False

Q65. _____ holds data, instructions, and information for future use.

- (a) Input device
- (b) Output device
- (c) Storage device
- (d) Communication device

Q66. What are the network classifications?

- (a) Personal Area Network
- (b) Local Area Network
- (c) Metropolitan Area Network
- (d) All the above

- Q67. The Wide Area Network is used for _____.
- (a) short-range
 - (b) building/campus
 - (c) community
 - (d) greater distances
- Q68. A network is a collection of computers and devices connected together via only a wiring communication device.
- (a) True
 - (b) False
- Q69. One of the basic network topologies is _____.
- (a) star
 - (b) bus
 - (c) ring
 - (d) any of the above
- Q70. The network protocols are _____.
- a. utility program.
 - b. Java protocol.
 - c. rules by which activities are conducted on a network.
 - d. network topologies.
- Q71. An example of the network protocols is _____.
- (a) HTTP Hyper Text Transfer Protocol
 - (b) TCP Transmission Control Protocol
 - (c) IP Internet Protocol
 - (d) any of the above
- Q72. The Internet service(s) is(are) _____
- (a) Voice over IP
 - (b) E-mail
 - (c) Virtual Private Networks
 - (d) All the above
- Q73. Building a large network from smaller ones can be done by using _____.
- (a) repeaters
 - (b) switches
 - (c) routers
 - (d) all the above
- Q74. Which of the following devices must you use to connect your computer to the Internet?
- (a) Sound card
 - (b) Scanner
 - (c) TV set
 - (d) Router or modem

Q75. Client-server network means server must be prepared to serve only three clients at any time.

- (a) True
- (b) False

Q76. Peer-to-peer (P2P) network means peers communicate as equals on a one-to-one basis.

- (a) True
- (b) False

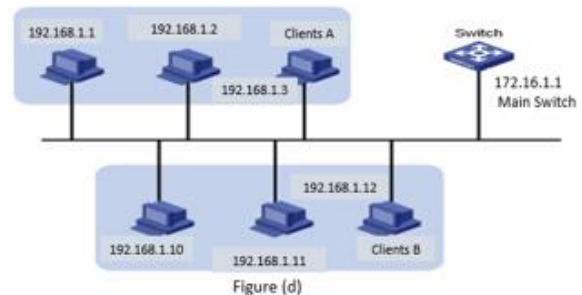
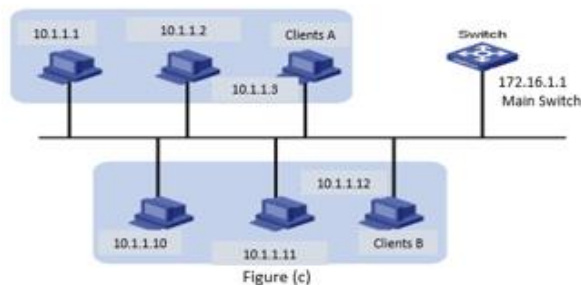
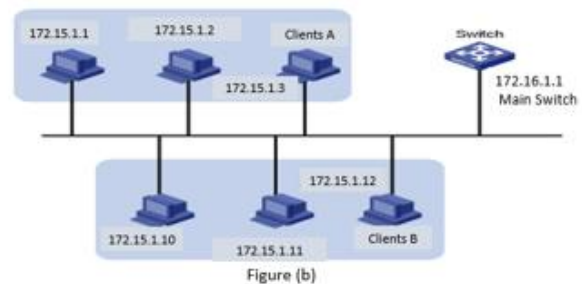
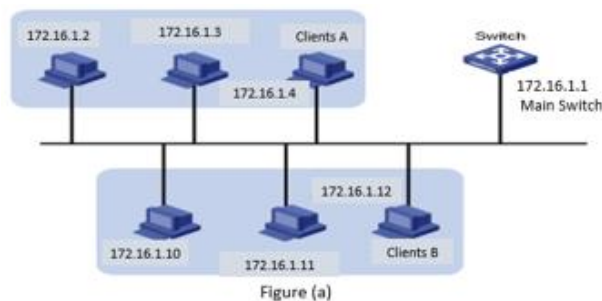
Q77. The Internet is a collection of _____.

- (a) networks
- (b) devices
- (c) agencies
- (d) computers

Q78. An IP address is _____.

- (a) a text address.
- (b) a numerical address
- (c) a mix of numbers and text address
- (d) a combination of an ASCII code

Q79. If your home network is shown in the next Figure with the IP's setting, choose the right Figure which represents the correct IP setting to each client of that network.



- (a) Figure (a)
- (b) Figure (b)
- (c) Figure (c)
- (d) Figure (d)

Q80. An example of client-server architectures is _____.

- (a) One-Tier Architecture
- (b) Two-Tier Architecture
- (c) Three-Tier Architecture
- (d) Any of the above

Q81. If your home network of class “A” has three clients, what are the expected IP addresses of those clients?

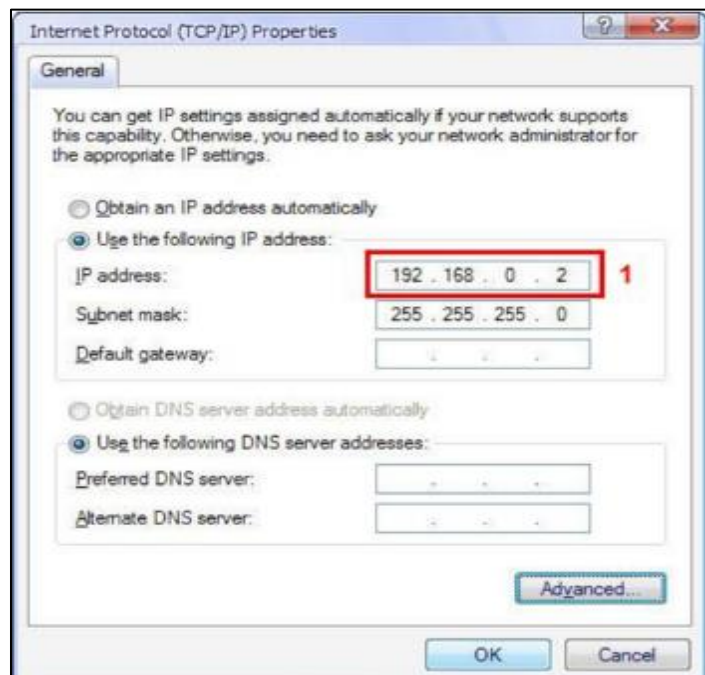
- (a) 11.1.1.2 - 11.1.1.3 - 11.1.1.4
- (b) 100.1.1.2 - 100.1.1.3 - 100.1.1.4
- (c) 10.1.1.2 - 10.1.1.3 - 10.1.1.4
- (d) 110.1.1.2 - 110.1.1.3 - 110.1.1.4

Q82. By using the Transmission Control Protocol/ Internet Protocol (TCP/IP), the network administrator can assign a unique IP address to each client within his/her home network.

- (a) True
- (b) False

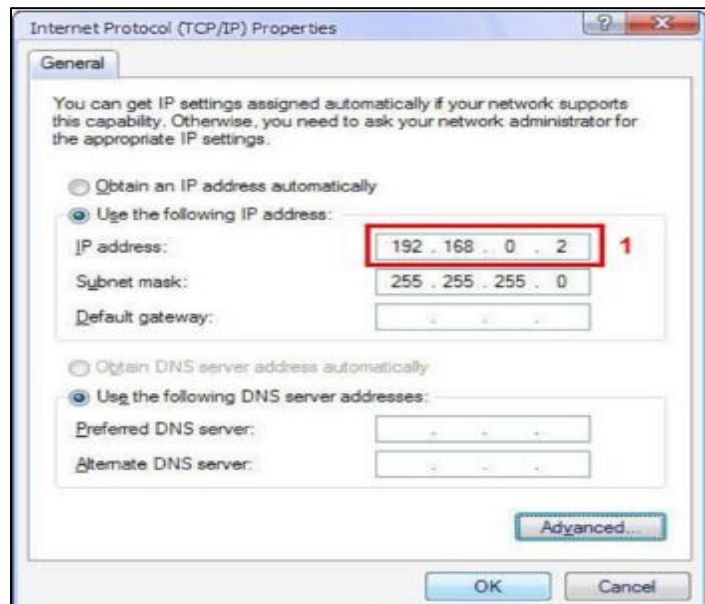
Q83. The next Figure demonstrates the IP setting of your network. What is the IP class of the network shown in the next Figure? (Hint: Class D is not applicable.)

- (a) Class A
- (b) Class B
- (c) Class D
- (d) Class C



Q84. The next Figure demonstrates the IP setting of your home network. What is the subnet mask of the home network shown in the next Figure?

- (a) 255.0.0.0
- (b) 255.255.255.0
- (c) None of the above
- (d) 255.255.0.0



Q85. An example of a private IP address is like _____.

- (a) 19.168.10.5
- (b) 192.167.10.5
- (c) 11.168.10.5
- (d) 172.15.10.5

Q86. An example of a public IP address is like _____.

- (a) 10.100.10.5
- (b) 192.168.10.5
- (c) 172.16.10.5
- (d) 11.168.10.5

Q87. An Internet Protocol (IP) address is not a unique numerical identifier for every device or network that connects to the internet.

- (a) True
- (b) False

Q88. The static IP address is manually assigned by the user or network administrator.

- (a) True
- (b) False

Q89. The dynamic IP address is automatically assigned by Dynamic Host Configuration Protocol DHCP server.

- (a) True
- (b) False

Q90. The IP address consists of 4 blocks separated by dot, and each block consists of 8 bits. It means that the IP address contains _____ bits.

- (a) 8
- (b) 16
- (c) 32
- (d) 64

Q91. The private cryptography system consists of _____key(s).

- (a) one
- (b) two
- (c) three
- (d) four

Q92. Where will you go to get an Internet connection for your network?

- a. Software development company
- b. Hardware company
- c. Internet café
- d. Internet Service Provider ISP

Q93. If your home network of class “A” has three clients, what are the expected IP addresses of those clients?

- a. 11.1.1.2 - 11.1.1.3 - 11.1.1.4
- b. 100.1.1.2 - 100.1.1.3 - 100.1.1.4
- c. 10.1.1.2 - 10.1.1.3 - 10.1.1.4
- d. 110.1.1.2 - 110.1.1.3 - 110.1.1.4

Q94. By using the Transmission Control Protocol/ Internet Protocol (TCP/IP), the network administrator can assign a unique IP address to each client within his/her network.

- a. True
- b. False

Q95. The Most common types of malware attacks are _____

- a. Viruses.
- b. Worms.
- c. Trojans.
- d. All the above

Q96. The number of hosts per network of class "A" is _____.

- a. 16,777,214
- b. 65,534
- c. 254
- d. 126

Q97. The number of networks of class "A" is _____.

- a. 16,384
- b. 2,097,152
- c. 126
- d. 254

Q98. The number of hosts per network of class "B" is _____.

- a. 16,777,214
- b. 65,534
- c. 254
- d. 126

Q99. If your home network of class "B" has three clients, what are the expected IP addresses of those clients?

- a. 172.15.1.2 - 172.15.1.3 - 172.15.1.4
- b. 170.16.1.2 - 170.16.1.3 - 170.16.1.4
- c. 172.30.1.2 - 172.30.1.3 - 172.30.1.4
- d. 172.31.1.2 - 172.31.1.3 - 172.30.1.4

Q100. The Internet is a worldwide collection of _____.

- a. servers
- b. agencies
- c. networks
- d. database

Q101 The World Wide Web consists of a worldwide collection of _____

- (a) electronic devices
- (b) electronic numbers
- (c) IP's addresses
- (a) electronic documents

Q102. The function(s) of a Web server _____.

- (a) delivers requested Web pages to the computers of the end users.
- (b) manages databases for the organizations.
- (c) manages application programs for educational purposes.
- (d) manages information systems inside the companies.

Q103. Which of the following is not a well-known Web browser?

- (a) Internet Explorer
- (b) Windows Explorer
- (c) Safari
- (d) Google Chrome

Q104. _____ is a web browser.

- (a) Internet Expo
- (b) Firebox
- (c) Safar
- (d) Opera

Q105. The server is _____.

- (a) an application program.
- (b) a computer.
- (c) an operating system.
- (d) a utility program.

Q106. Which language is used to create a web page?

- (a) Wireless Markup Language WLM
- (b) Hypertext Markup Language HTML
- (c) Hypertext Transfer Language HTL
- (d) Standard General Language SGL

Q107. Google is the search engine on the Web, handling about _____% of all Web searches.

- (a) 17
- (b) 13
- (c) 56
- (d) 4

Q108. Virtual reality is an example of _____.

- (a) non-interactive computer graphics.
- (b) 3D environment simulation.
- (c) image processing.
- (d) screen saver.

Q109. The HTML is _____ language.

- (a) machine
- (b) object-oriented
- (c) tags
- (d) logical

Q110. Which of these elements are all <table> elements?

- (a) <table><head><tfoot>
- (b) <table><tr><td>
- (c) <table><body><tr>
- (d) <table><tr><tt>

Q111. What is the correct HTML for adding a background color?

- (a) <body bg="yellow">
- (b) <body style="background-color: yellow;">
- (c) <background>yellow</background>
- (d) < background ="yellow">

Q112. Choose the correct HTML element to define important text_____.

- (a) <important>
- (b)
- (c) <i>
- (d)

Q113. HTML element to define emphasized text _____.

- (a) <i>
- (b)
- (c) <italic>
- (d) <I>

Q114. HTML for creating a hyperlink is _____

- (a) <a> http://www.msa.edu.eg
- (b) MSA University
- (c) MSA University
- (d) MSA University

Q115. HTML element to make a numbered list _____.

- (a)
- (b) <list>
- (c) <dl>
- (d)

Q116. The syntax of inserting image on the webpage using HTML.

- (a)
- (b)
- (c)
- (d) <image href="url address" alt="alternate text">

Q117. What is the correct HTML element for playing video files?

- (a) <media>
- (b) <video>
- (c) <movie>
- (d) <cinema>

Q118. Audio does not include_____

- (a) music
- (b) speech
- (c) any sounds
- (d) images

Q119. E-commerce is a business transaction over_____

- (a) the electronic network
- (b) the Internet
- (c) the social networks
- (d) any of the above

Q120. The factor(s) that affect the E-commerce_____

- (a) political
- (b) sports
- (c) management
- (d) international competitions

Q121. Forward auction in which the buyer is asking for bids from sellers, and the_____.

- (a) lowest bid wins.
- (b) largest bid wins.
- (c) auction coordinator wins.
- (d) buyer loses.

Q122. Shoppers like Wal-Mart online or Amazon.com are examples for business model _____

- (a) B2C
- (b) B2B
- (c) C2B
- (d) C2C

Q123. Tells the computer what tasks to perform and how to perform them _____.

- (a) Wi-Fi
- (b) Software
- (c) Router
- (d) Computer

Q124. What are the categories of the application software?

- (a) Business - Communications
- (b) Graphics and Multimedia
- (c) Home/Personal/Educational
- (d) All the above

Q125. What information do you need to send an electronic message to your friend?

- (a) Telephone
- (b) Home address
- (c) E-mail address
- (d) Company address

Q126. What is(are) application software?

- (a) Packaged software
- (b) Open-source software
- (c) Shareware software
- (d) All the above

Q127. Uploading is the process of _____ information.

- (a) receiving
- (b) transmitting
- (c) analyzing
- (d) writing

Q128. Kotlin is the best programming language for building mobile apps because_____

- (a) simple coding and used for the android platform
- (b) easy in maintenance
- (c) completely compatible with Java
- (d) all the above

Q129. Swift could be the best mobile apps programming language because_____

- (a) completely compatible with Java
- (b) requires fewer lines of code and quick development.
- (c) it works only with the android platform.
- (d) developed by Google

Q130. A Web page has a unique address called _____ address

- (a) URL
- (b) Web
- (c) Text
- (d) Any of the above

Q131. Which of the following type of devices is a computer microphone?

- (a) Output device
- (b) Speaker device
- (c) Keyboard device
- (d) Input device

Q132. The main circuit board of the computer is called_____

- (a) ALU
- (b) CPU
- (c) Mother board
- (d) System unit

Q133. The component of the processor that performs mathematical calculation is _____

- (a) Control Unit
- (b) ROM
- (c) ALU
- (d) CPU

Q134. Which of the following ports can provide Internet connection to the computer?

- (a) Ethernet
- (b) USB
- (c) HDMI
- (d) VEGA

Q135. Which one of these facts are false?

- (a) RAM can't store permanent data
- (b) ROM is memory that cannot be changed by a program or user.
- (c) ROM is embedded, can't be removed.
- (d) If a computer loses power, all the data is saved onto the RAM.

Q136. Monitor is called_____

- (a) Output device
- (b) Input device
- (c) Scanner device
- (d) Storage device

Q137. You can use this device to input printed documents to a computer?

- (a) Tracker ball
- (b) Keyboard
- (c) Mouse
- (d) Scanner

Q138. Convert an ASCII = Bc4 to Binary = _____

- (a) 010000110110001100110101
- (b) 010000100110001100110100
- (c) 010000010110000100110110
- (d) 010000110110001000110111

Q139. 4 in ASCII = _____ and in binary = _____

- (a) 53 - 00110101
- (b) 48 - 00110000
- (c) 52 – 00110100
- (d) 50 – 00110010

Q140. The system clock controls the timing of all computer operations.

- (a) False
- (b) True

Q141. Plotters are used to produce high-quality drawings such as drawing of size A0 or called A node.

- (a) True
- (b) False

Q142. Convert Binary = 010011010100011101100111 to ASCII = _____

- (a) GMg
- (b) gMG
- (c) GMg
- (d) MGg

Q143. Giga Byte = _____

- (a) 1024 Byte
- (b) 1000 Mega Byte
- (c) 1024 Mega Byte
- (d) 1000 K Byte

Q144. Radio Frequency Identification RFID Tracking Systems are used in _____

- (a) highway Hi-pass
- (b) logistics Management System
- (c) transportation Card
- (d) all the above

Q145. Each class "C" address has a number of hosts _____

- (a) 16,777,214
- (b) 65,534
- (c) 254
- (d) 126

Q146. An example of the MAC addresses is _____.

- (a) 00: A0:C9:14:C8:29
- (b) 192.168.1.1
- (c) www.msa.edu.eg
- (d) 14C829

Q147. A program that controls a computer's basic functions is called _____.

- (a) Hard Drive
- (b) Motherboard
- (c) Operating System
- (d) CPU

Q148. Multimedia refers to any application that combines text with _____

- (a) graphics
- (b) animation
- (c) virtual Reality
- (d) any of the above

Q149. The best use of augmented reality in _____

- (a) business
- (b) education
- (c) sport
- (d) industry

Q150. The World Wide Web consists of a worldwide collection of _____

- (a) electronic devices
- (b) electronic numbers
- (c) the IP's addresses.
- (d) electronic documents